

Midterm 1 Practice — Lectures 1–11

M 375T Mathematical Foundations of Machine Learning — Fall 2026

Midterm Tuesday, October 6, in class · 75 minutes · closed book, no devices · four questions, 10 points each

NumPy you should be able to read:

`np.arange`, `reshape`, `np.array`, `.sum(axis=0)`, `.mean(axis=1)`, `x.mean()`, `x.std()`, `x.var()`,
`np.cov`, `np.column_stack`, `rng.normal`, `rng.uniform`, `(cond).sum(axis=1)`, `np.mean(cond)`,
`np.abs`, `np.log`, `np.exp`, `np.linspace`, `np.argmax`, `np.round`, `W.sum(axis=1)`, `W / d[:, None]`,
`P.T @ mu`, `P @ P`, `np.ones`, `np.diag`, `np.eye`, `v @ L @ v`, `np.linalg.eigvalsh` (eigenvalues of a
symmetric matrix, increasing order), `np.linalg.eigvals`.

Conventions. $P_{ij} = W_{ij}/d_i$ (rows sum to one); laws are column vectors, $\mu_{t+1} = P^T \mu_t$; $Q = \frac{1}{2}(I + P)$; $L = D - W$. `rng = np.random.default_rng(0)` throughout. Do everything on paper first, then check in NumPy.

Part A — Samples, distributions, and NumPy

A1. Shapes and axes. Predict the four outputs. Do not run the code.

```
z = np.arange(6).reshape(3, 2)
print(z.shape)                # line 1
print(z.sum(axis=0))          # line 2
print(z.mean(axis=1))         # line 3
print((z > 2).sum())          # line 4
```

Say in one sentence what `axis=0` and `axis=1` each collapse, and why line 4 counts something.

A2. The uniform on $[0, 4]$. Let X be uniform on $[0, 4]$.

- (a) Write the density $f(x)$ and compute by integration $\mathbb{E}[X]$, $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$, and $\mathbb{P}(X \in [1, 2])$.
- (b) The code below estimates the same three numbers. Write the value each line approaches as the sample size grows.

```
u = rng.uniform(0, 4, size=500_000)
print(u.mean())
print(u.var())
print(np.mean((u >= 1) & (u <= 2)))
```

A3. Probabilities are averages of indicators.

```
Z = rng.normal(size=100_000)
X = 3 + 2*Z
print(np.mean(X > 3))          # line 1
print(np.mean(X > 5))          # line 2

n = 100_000
Pts = np.column_stack([rng.uniform(-2, 2, n), rng.uniform(-1, 1, n)])
```

```
inside = Pts[:, 0]**2 / 4 + Pts[:, 1]**2 <= 1
print(8 * inside.mean())          # line 3
```

- (a) Predict line 1 exactly, and line 2 as a probability of a standard normal (you may quote $\mathbb{P}(Z > 1) \approx 0.1587$). What does `np.mean` of a boolean array compute?
- (b) Line 3 estimates the area of the ellipse $x^2/4 + y^2 \leq 1$. Explain the factor 8, and give the value the line approaches.

Part B — Gaussians and maximum likelihood

B1. Standardize. A feature has sample mean $\hat{\mu} = 6$ and sample standard deviation $\hat{\sigma} = 2$. We standardize it: $z = (x - \hat{\mu})/\hat{\sigma}$. (i) What value does the data point $x = 8$ map to? (ii) Which original value x maps to $z = -1.5$? (iii) After standardizing, what is the only Gaussian candidate left, and why?

B2. Maximum likelihood for a Gaussian, from code.

```
x = np.array([1., 3., 5., 7., 9.])
sig = 1.0
ms = np.linspace(0, 10, 201)
H = [np.mean(0.5*np.log(2*np.pi*sig**2) + (x - m)**2 / (2*sig**2))
      for m in ms]
print(ms[np.argmin(H)])          # line 1
print(min(H))                    # line 2
z = (x - x.mean()) / x.std()
print(z)                         # line 3
print(np.mean(-np.log(np.exp(-z**2/2) / np.sqrt(2*np.pi)))) # line 4
```

- (a) Write $H(m)$ as a formula and name it. Predict line 1 by computing $dH/dm = 0$. Would the answer change for `sig = 3.0`?
- (b) Predict line 2 as an exact expression: substitute $m = \bar{x}$ and use $\frac{1}{n} \sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$ (compute $\hat{\sigma}^2$ for these five numbers).
- (c) Predict lines 3 and 4. Why is line 4 equal to $\frac{1}{2} \log(2\pi) + \frac{1}{2} \approx 1.419$ for every dataset? What does that say about the cross-entropy score as a judge of shape?
- (d) Now let both m and σ vary. Show that the minimizer of $H(m, \sigma) = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \cdot \frac{1}{n} \sum_i (x_i - m)^2$ is $m = \bar{x}$, $\sigma^2 = \hat{\sigma}^2$, and that the minimum value is $\frac{1}{2} \log(2\pi e \hat{\sigma}^2)$.

B3. One dial left: the correlation. For standardized 2D data ($\overline{x_1} = \overline{x_2} = 0$, $\overline{x_1^2} = \overline{x_2^2} = 1$) with empirical correlation $c = \overline{x_1 x_2}$, the cross-entropy against $N(0, \Sigma_\rho)$, $\Sigma_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, is

$$H(\rho) = \log(2\pi) + \frac{1}{2} \log(1 - \rho^2) + \frac{1 - \rho c}{1 - \rho^2}.$$

- (a) The code below evaluates H on a grid and prints the minimizer. For $c = 0.6$, what does it print (to two decimals), and why? (b) Show by differentiating that the only critical point of H in $(-1, 1)$ is $\rho = c$.

```

c = 0.6
rhos = np.linspace(-0.99, 0.99, 199)
H = np.log(2*np.pi) + 0.5*np.log(1 - rhos**2) + (1 - rhos*c) / (1 - rhos**2)
print(round(rhos[np.argmin(H)], 2))

```

B4. Any matrix is a Gaussian factory. Each row of Z is a pair of independent standard normals.

```

A = np.array([[2., 0.],
              [1., 1.]])
Z = rng.normal(size=(200_000, 2))
X = Z @ A.T
print(np.cov(X.T).round(1))          # line 1

S = np.array([[2., 1.],
              [1., 2.]])
print(np.linalg.eigvalsh(S))         # line 2
print(S @ np.array([1., 1.]))       # line 3

```

- Each row of X is a sample of $N(0, \Sigma)$ with $\Sigma = AA^T$. Compute Σ and predict line 1.
- Predict lines 2 and 3. Give both eigenvectors of S .
- A cloud sampled from $N(0, S)$ is an ellipse of points. In which direction is it longest, and what is the standard deviation of the cloud along that direction?
- Is A the only matrix with $AA^T = \Sigma$? Name one other way to build a matrix B with $BB^T = S$ from the eigenvalues and eigenvectors of S .

Part C — Relative entropy, Sanov, and the two-bin test

C1. Four bins, Sanov. Quarter the interval into four bins. The model is uniform, $u = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, and $n = 100$ samples produce the observed histogram $q = (\frac{1}{2}, \frac{1}{2}, 0, 0)$.

- Compute $D(q||u) = \sum_{\omega} q(\omega) \log \frac{q(\omega)}{u(\omega)}$ exactly (recall $0 \log 0 = 0$).
- Using Sanov's estimate $\mathbb{P}(\text{histogram} \approx q) \approx e^{-nD(q||u)}$, write this probability as a power of 2. Should we keep the uniform model?
- The line `np.sum(q * np.log(q / u))` prints `nan` for this q . Why, and how do you fix it?

C2. A coin in disguise. Each entry of X is an independent $N(0, 1)$ sample. Predict the first output exactly; for the other two, give the values they approach as the number of rows grows.

```

import numpy as np
rng = np.random.default_rng(42)
X = rng.normal(size=(10_000, 100))
q = (X > 0).mean(axis=1)
print(q.shape)
print(q.mean())
print(q.std())

```

Then: in 100 coin flips, 60 are heads. The code uses $D(\hat{q}||p) \approx 2\epsilon^2$ with $\epsilon = k/n - \frac{1}{2}$, $p = (\frac{1}{2}, \frac{1}{2})$, and rejects when $2nD > 3.84$. Write all four outputs.

```
n, k = 100, 60
eps = k/n - 0.5
D = 2 * eps**2
T = 2 * n * D
print(round(eps, 2))
print(round(D, 2))
print(round(T, 2))
print(T > 3.84)
```

C3. Is the model right? Each row of X is one data set of $n = 400$ independent samples from the uniform distribution on $[-2, 2]$.

```
rng = np.random.default_rng(0)
X = rng.uniform(-2, 2, size=(50_000, 400))
k = (np.abs(X) < 1).sum(axis=1)
print(k.shape)                # line 1
print(k.mean() / 400)         # line 2
print(k.std())                # line 3
eps = k / 400 - 0.5
D = 2 * eps**2
print(np.mean(2 * 400 * D > 3.84))    # line 4
```

- Predict line 1. Say in one sentence what one entry of k counts.
- Predict line 2 (the value it is close to). Justify it with the probability that one sample lands in $(-1, 1)$.
- Predict line 3 (the value it is close to). Name the distribution of one entry of k and use its variance.
- Predict line 4 (the value it is close to). Which theorem says so, and what is the role of the number 3.84?
- A different experiment produces one data set of 400 samples from an unknown distribution, and 170 of them land in $(-1, 1)$. Compute ϵ , D and $2nD$ for this data set. Is it consistent with the uniform distribution on $[-2, 2]$?

Part D — Graphs, random walks, and the Markov property

D1. A walk from code.

```
import numpy as np
W = np.array([[0, 1, 2, 0],
              [1, 0, 1, 0],
              [2, 1, 0, 1],
              [0, 0, 1, 0]], dtype=float)
d = W.sum(axis=1)
P = W / d[:, None]
```

```

mu = np.array([0., 0., 0., 1.])
for t in range(2):
    mu = P.T @ mu
print(mu)                                # line 1
print((P @ P)[3, 3])                    # line 2

u = np.ones(4) / 4
print(P.T @ u)                          # line 3

```

- Write d and P .
- Predict line 1. Say in one sentence what the vector μ is at the moment it is printed.
- Predict line 2. Which routes of the walker does this number add up?
- Write, from memory, the equation an invariant measure π satisfies, and find π for this graph. Then predict entry 3 of line 3 (the last entry) and say what it shows about the uniform vector u .

D2. A weighted path. Vertices 0, 1, 2; the only edges are 0–1 with weight 2 and 1–2 with weight 1.

- Write W , d , P , and $Q = \frac{1}{2}(I + P)$.
- Predict the output of the code. Do not run it.

```

mu = np.array([1., 0., 0.])
for t in range(3):
    mu = P.T @ mu
    print(mu)
print((P @ P)[0, 0])
print(Q[1])

```

- Explain in one sentence what the number $(P @ P)[0, 0]$ is, as a probability. List the routes that contribute to it.
- Explain why $P @ \mu$ would be wrong for moving a law. Give the check that catches the mistake.

D3. The star. Vertices 0, 1, 2, 3. Vertex 0 is joined to each of 1, 2, 3 with weight 1; there are no other edges.

- Write d , P , π .
- Start at vertex 1. Write the ordinary laws $\mu_0, \mu_1, \mu_2, \mu_3$. Which two laws alternate? Why does the ordinary law never converge on this graph?
- Write the lazy laws for $t = 0, 1, 2$. What do they converge to?
- Give an eigenvector of P with eigenvalue -1 . Check it by multiplying. What is its eigenvalue for Q ?

D4. Walking on a graph. The code builds the weight matrix of an undirected graph on the vertices 0, 1, 2, 3. The walker moves with the rule $P_{ij} = W_{ij}/d_i$.

```
import numpy as np
W = np.array([[0, 2, 0, 1],
              [2, 0, 1, 0],
              [0, 1, 0, 1],
              [1, 0, 1, 0]])
```

- Draw the graph: the four vertices, every edge, and each edge's weight.
- Write the degree vector d and the transition matrix P .
- The walker starts at vertex 0, so $\mu_0 = (1, 0, 0, 0)^T$. Give μ_1 , the law after one step.
- Write down, from memory, the equation that an invariant measure π of the walk satisfies. Find π for this graph; it must be a probability vector. Verify your equation at vertex 0 by explicit computation. In one sentence: what does π describe about a long walk?

Part E — The invariant measure and eigenvalues

E1. Fixed point of the update. Graph of D2.

- Write down, from memory, the equation for an invariant measure and the two conditions that make it a probability vector.
- Find π . Verify the equation at every vertex by explicit arithmetic.
- Prove that $Q^T \pi = \pi$ follows from $P^T \pi = \pi$.
- Predict the output of `u = np.ones(3) / 3; print(P.T @ u)`. Is u invariant? In one sentence: why must vertex 1 carry more mass than vertex 2 in equilibrium?

E2. Which are eigenvectors? Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad w = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Which of the vectors v, w, u are eigenvectors of A ? For each one that is, give its eigenvalue. Justify each answer with a computation (no characteristic polynomial needed). Then: $A = I + \mathbf{1}\mathbf{1}^T$ is symmetric; what does the spectral theorem say about its eigenvalues and eigenvectors, and how many eigenvectors with eigenvalue 1 are there?

E3. Eigenvalues without symmetry.

- The shear $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and the matrix $B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. For each: find the eigenvalues, find all eigenvectors, and say whether the matrix is diagonalizable. Which one does the spectral theorem apply to, and why?
- The directed 3-cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ has $P_{01} = P_{12} = P_{20} = 1$ and all other entries 0. Predict the output of the code and explain why the ordinary law never converges although the lazy law does.

```
P = np.array([[0, 1, 0],
              [0, 0, 1],
```

```

[1, 0, 0.]]
Q = (np.eye(3) + P) / 2
mu = np.array([1., 0., 0.])
nu = np.array([1., 0., 0.])
for t in range(3):
    mu = P.T @ mu
    nu = Q.T @ nu
print(mu)
print(np.round(nu, 3))
print(np.round(np.abs(np.linalg.eigvals(P)), 3))

```

- (c) Why does $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ satisfy $P^T \pi = \pi$ here, although W is not symmetric and $\pi \propto d$ is not a theorem for directed graphs?

Part F — The Laplacian, energy, and convergence of the lazy walk

F1. Two triangles and a bridge. Vertices $0, \dots, 5$; edges of weight 1 inside each triangle $\{0, 1, 2\}$ and $\{3, 4, 5\}$; one bridge $2-3$ of weight 1. W is its weight matrix.

```

d = W.sum(axis=1)
L = np.diag(d) - W
v = np.array([1., 1., 1., -1., -1., -1.])
print(v @ L @ v)           # line 1
print(L @ v)               # line 2
print(np.linalg.eigvalsh(L)[0]) # line 3

```

- Write d , π , and row 2 of P and of Q .
- Predict line 1 with the Dirichlet form. Which edges contribute?
- Predict line 2. Is v an eigenvector of L ? Which entries break it, and why are the bridge vertices the problem?
- Predict line 3 and say why. Check that $(1, -1, 0, 0, 0, 0)^T$ is an eigenvector of L and give its eigenvalue.
- Compute $\text{cut}(A, B)$, $\text{vol}(A)$, $\text{vol}(B)$ and $\text{Ncut}(A, B) = \text{cut}(A, B) \left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)} \right)$ for $A = \{0, 1, 2\}$, $B = \{3, 4, 5\}$. In two sentences: why does the sign of v_2 (the second eigenvector of $Lv = \lambda Dv$) recover the two triangles?

F2. The lazy walk on the 4-cycle, from code. The graph is the unweighted 4-cycle $0-1-2-3-0$, all weights 1.

```

W = np.array([[0, 1, 0, 1],
              [1, 0, 1, 0],
              [0, 1, 0, 1],
              [1, 0, 1, 0]], dtype=float)
d = W.sum(axis=1)
L = np.diag(d) - W
P = W / d[:, None]
Q = (np.eye(4) + P) / 2

```

```

print(np.linalg.eigvalsh(L))      # line 1
print(np.linalg.eigvalsh(Q))      # line 2
mu = np.array([1., 0., 0., 0.])
for t in range(3):
    mu = Q.T @ mu
print(np.round(mu, 3))           # line 3

```

- (a) Check that $(1, 1, 1, 1)$, $(1, 0, -1, 0)$, $(0, 1, 0, -1)$, $(1, -1, 1, -1)$ are eigenvectors of L and compute their eigenvalues. Predict line 1.
- (b) Why is Q symmetric for this graph, so that `eigvalsh` is allowed? Write $Q = I - cL$ with the right constant c and predict line 2.
- (c) Write e_0 as a combination of the four eigenvectors, write μ_t as a formula in t , and predict line 3.
- (d) Which eigenvector's term is gone after a single step, and what does that say about the ordinary walk on this graph?

F3. Why the lazy walk converges. Using the expansion $\mu_t = \pi + \sum_{k \geq 2} a_k \eta_k^t w_k$: state which two facts about the η_k are needed for $\mu_t \rightarrow \pi$, and say which property of the graph gives each fact (connectedness; laziness). Then explain in one sentence why $a_1 w_1 = \pi$ for every starting law μ_0 .

Solutions will be posted after Thursday's class. Bring questions to office hours (Tue/Thu 5–6 PM, PMA 9.156).